Analytical Questions

Analytical Queries

1. Calculate the total premium collected during the year 2024.

Ans:

```sql
-- Q1:
select sum(premium) AS premium_2024 from policy_sales
where year(Policy_Purchase_Date) = 2024
```

102 %

Results | Messages

	premium_2024
1	240039700

Total Premium Collected

₹ 24,00,39,700

All the policies are sold in 2024 so we can validate from Power BI the results of SQL. Also if multiple year policies are present in the data we can use slicer for that year.

2. Calculate the total claim cost for each year (2025 and 2026) with a monthly breakdown.

Ans:

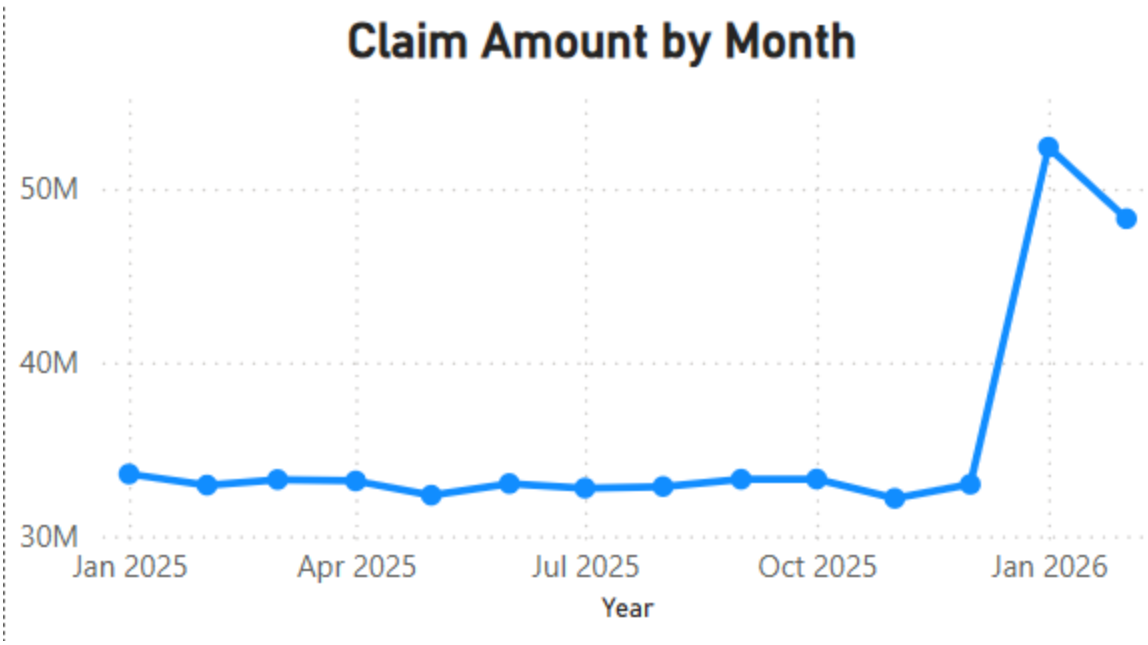
```sql
-- Q2:
select year(Claim_date) as claim_year ,
month(Claim_date) as claim_month ,
sum(Claim_amount) as claim_amount
from claims
where year(Claim_date) in (2025,2026)
group by year(Claim_date) , month(Claim_date)
order by year(Claim_date) , month(Claim_date)
```

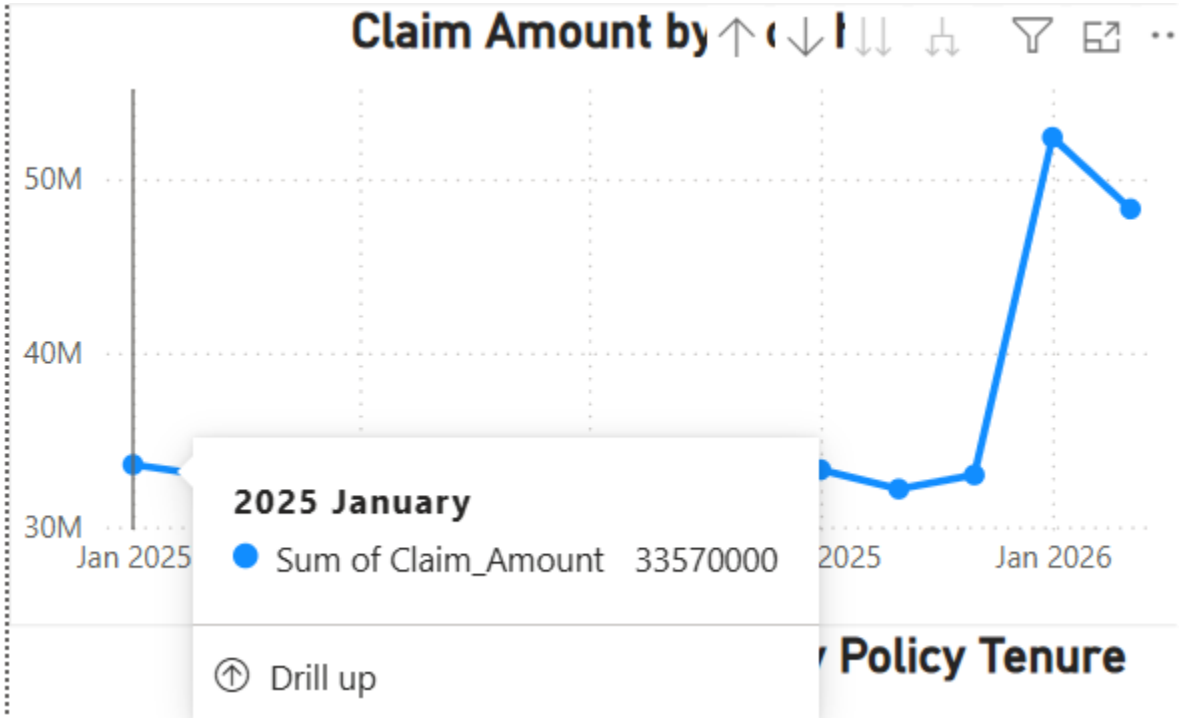
102 %

Results | Messages

	claim_year	claim_month	claim_amount
1	2025	1	33570000
2	2025	2	32930000
3	2025	3	33250000
4	2025	4	33170000
5	2025	5	32350000
6	2025	6	33020000
7	2025	7	32760000
8	2025	8	32850000
9	2025	9	33270000
10	2025	10	33280000
11	2025	11	32170000
12	2025	12	32980000
13	2026	1	52400000
14	2026	2	48280000

We can validate this by aur power BI chart claim amount by month.


Claim Amount by Month
50M
40M
30M
Jan 2025
Apr 2025
Jul 2025
Oct 2025
Jan 2026
Year


Claim Amount by
50M
40M
30M
Jan 2025
2025
Jan 2026
2025 January
Sum of Claim_Amount 33570000
Drill up
Policy Tenure

3. Calculate the claim cost to premium ratio for each policy tenure (1, 2, 3, and 4 years).

Ans:

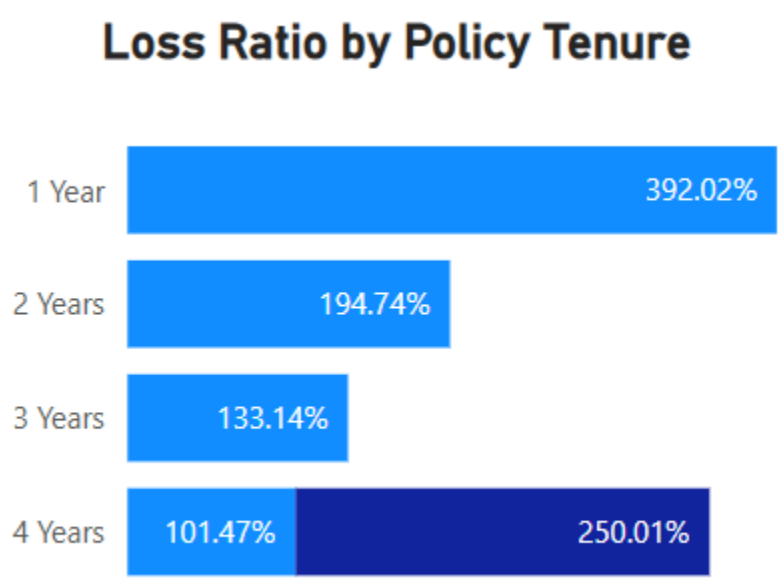
```sql
-- Q3:
select p.Policy_Tenure,
sum(c.Claim_Amount) as total_claims,
sum(p.Premium) as total_premium,
sum(c.Claim_Amount) * 1.0/ sum(p.Premium) as claim_cost_premium_ratio
from policy_sales p
LEFT JOIN claims c
on p.Vehicle_ID = c.Vehicle_ID
group by p.Policy_Tenure
order by p.Policy_Tenure;


--Q4:
```

102 %

Results | Messages

	Policy_Tenure	total_claims	total_premium	claim_cost_premium_ratio
1	1	78380000	19993800	3.920215266732
2	2	117000000	60080400	1.947390496734
3	3	159360000	119695500	1.331378372620
4	4	141540000	40432800	3.500623256366



4. Calculate the claim cost to premium ratio by the month in which the policy was sold (January–December 2024).

Ans:

```sql
--Q4:
select datename(month, p.Policy_Purchase_Date) as sale_month,
sum(c.Claim_Amount)* 1.0 / sum(p.Premium) as claim_ratio
from policy_sales p
left join claims c
on p.Vehicle_ID = c.Vehicle_ID
group by datename(month, p.Policy_Purchase_Date),month(p.Policy_Purchase_Date)
order by month(p.Policy_Purchase_Date);
```

102 %

Results | Messages

	sale_month	claim_ratio
1	January	2.078279231940
2	February	2.144617565237
3	March	2.055153560878
4	April	2.116047931730
5	May	2.015739810491
6	June	2.073125529959
7	July	2.026025849801
8	August	2.042431451969
9	September	2.095039536243
10	October	2.053804584805
11	November	2.067051013986
12	December	2.034656890891

We can validate this by simulation the month year slicer and checking the loss ratio KPI.

5. If every vehicle that has not yet made a claim eventually files exactly one claim during the remaining policy tenure, estimate the total potential claim liability.

Ans:

```sql
--Q5:
select count(*) as vehicles_without_claim,
cast(count(*) as bigint) * 10000 as potential_claim_liability
from policy_sales p
left join claims c
on p.Vehicle_ID = c.Vehicle_ID
where c.Vehicle_ID is null;
```

102 %

Results | Messages

	vehicles_without_claim	potential_claim_liability
1	950779	9507790000

6. Assume daily premium = Total Premium ÷ Total Policy Tenure Days. Based on this: • Calculate the premium already earned by the company up to February 28, 2026. • Estimate the premium expected to be earned monthly for the remaining policy period (assume 46 months remaining).

Ans:

```sql
-- Q6:
select
    sum(
        (Premium * 1.0 / datediff(day, Policy_Start_Date, Policy_End_Date)) *
        datediff(day, Policy_Start_Date, '2026-02-28')
    ) as earned_premium_till_feb_2026
from policy_sales
where Policy_Start_Date <= '2026-02-28';
```

102 %

Results | Messages

	earned_premium_till_feb_2026
1	66149276.712153471540

```sql
-- Q6:
select
    (sum(Premium) - (
        select
            sum(
                (p.Premium * 1.0 / datediff(day, p.Policy_Start_Date, p.Policy_End_Date)) *
                case
                    when '2026-02-28' > p.Policy_End_Date
                        then datediff(day, p.Policy_Start_Date, p.Policy_End_Date)
                    else datediff(day, p.Policy_Start_Date, '2026-02-28')
                end
            )
        from policy_sales p
        where p.Policy_Start_Date <= '2026-02-28'
    )) / 46.0 as expected_monthly_premium
from policy_sales;
```

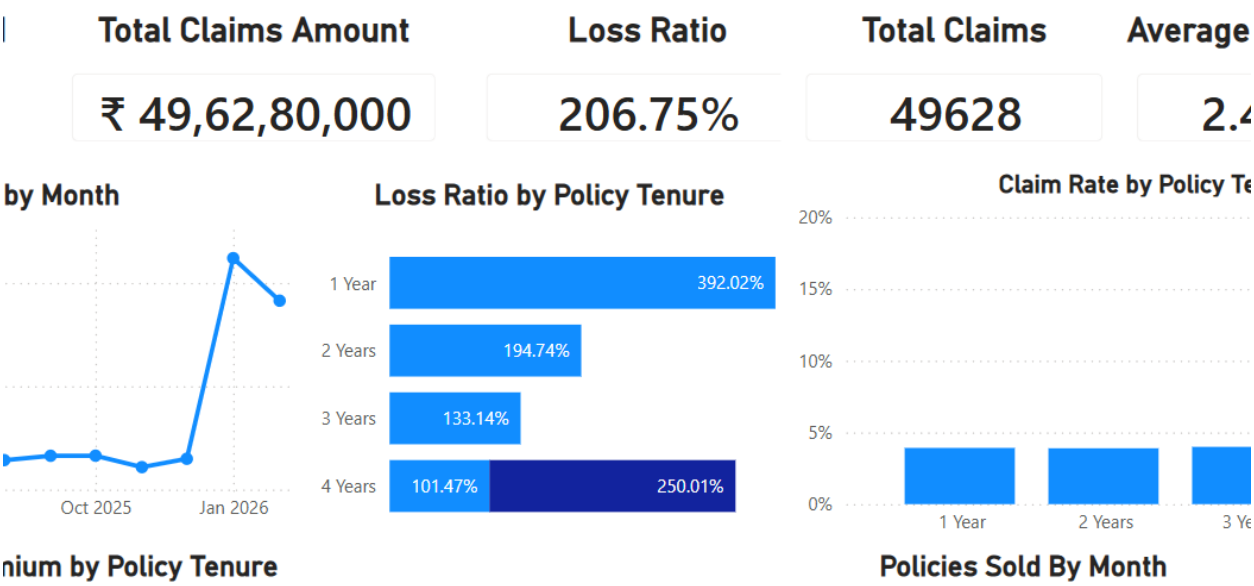
102 %

Results | Messages

	expected_monthly_premium
1	3786013.16260057923

7. Identify which policy tenure appears most profitable and explain why.

Ans:




```
-- Q3 Bonus
select p.Policy_Tenure, sum(c.Claim_Amount) as total_claims, sum(p.Premium) as total_premium, sum(c.Claim_Amount) * 1.0 / sum(p.Premium) as loss_ratio
from policy_sales p
left join claims c
on p.Vehicle_ID = c.Vehicle_ID
group by p.Policy_Tenure
order by p.Policy_Tenure;
```

The multi-part identifier "c.Vehicle_ID" could not be bound.

102 %

Results | Messages

	Policy_Tenure	total_claims	total_premium	loss_ratio
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The profitability of each policy tenure was evaluated using the loss ratio metric, calculated as total claims divided by total premium.

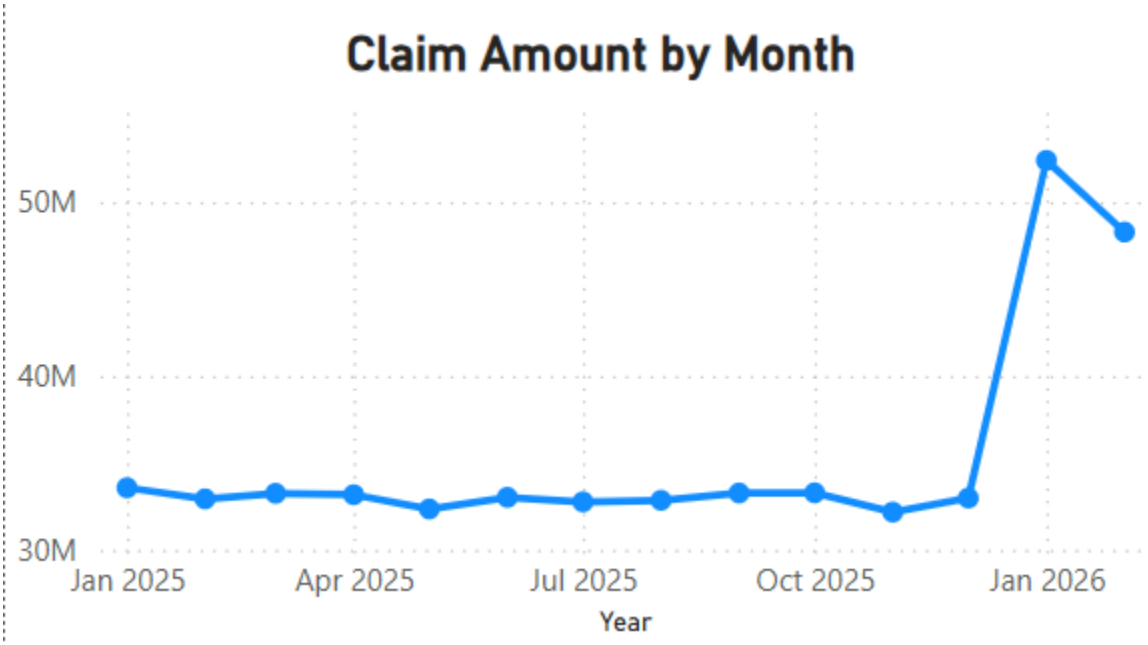
Based on the analysis, policies with a tenure of three years appear to be the most profitable. These policies have the lowest loss ratio among all tenure categories.

Shorter tenures such as one-year policies show significantly higher loss ratios because the premium collected is relatively small compared to the claim amount. As the policy tenure increases, more premium is collected, which improves the claim-to-premium balance.

Therefore, three-year policies provide the most favourable premium-to-claim relationship within the portfolio.

8. Build a simple dashboard or visualization showing claim trends by month.

Ans:



We can use Claims Count instead of claim amount but the graph will be same.

9. Estimate the loss ratio (Claims ÷ Premium) for the portfolio.

Ans:

```sql
-- Q3 Bonus
select p.Policy_Tenure, sum(c.Claim_Amount) as total_claims, sum(p.Premium) as total_premium, sum(c.Claim_Amount) * 1.0 / sum(p.Premium) as loss_ratio
from policy_sales p
left join claims c
on p.Vehicle_ID = c.Vehicle_ID
group by p.Policy_Tenure
order by p.Policy_Tenure;
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	Policy_Tenure	total_claims	total_premium	loss_ratio
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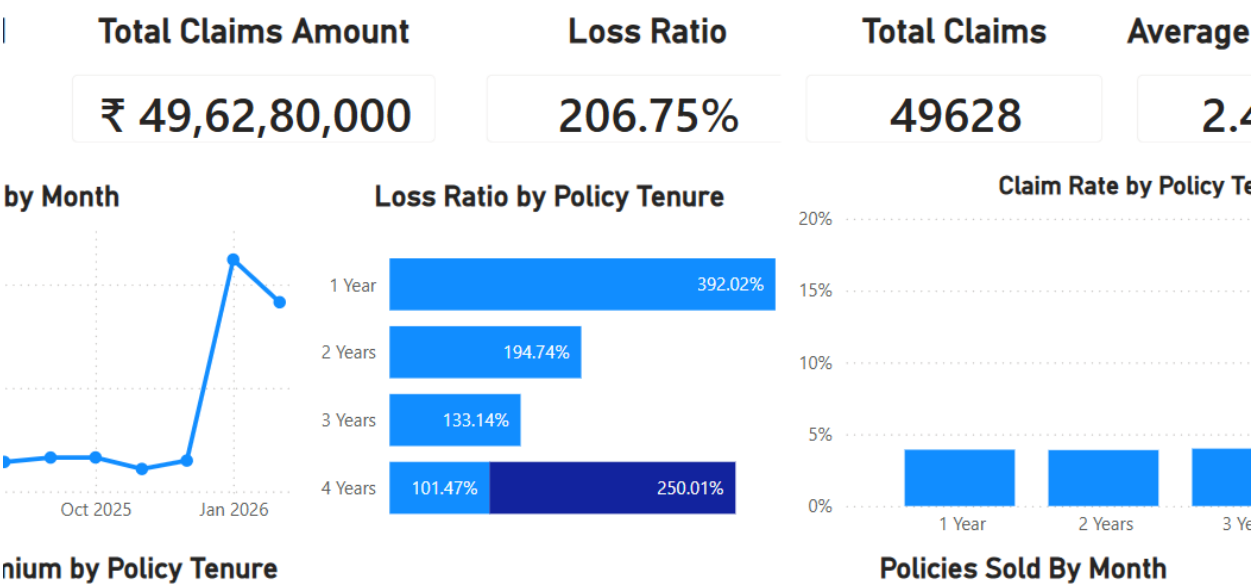

- Based on the analysis, policies with a tenure of three years appear to be the most profitable. These policies have the lowest loss ratio compared to other tenures.
- Shorter tenures, especially one-year policies, show a significantly higher loss ratio because the premium collected is relatively small compared to the fixed claim amount. As the policy tenure increases, the premium collected increases, which improves the loss ratio.
- Therefore, the three-year policy tenure provides the most balanced premium-to-claim relationship within the portfolio

10. If claim frequency increases by 5% annually, estimate the impact on future profitability.

Ans: Total Claims: Rs 4,96,28,000

If 5% increase then: Rs 4,96,28,000 + 5% of Rs 4,96,28,000 = Rs 5,21,09,400

Loss Ratio Old = Rs 4,96,28,000 / Rs 2,40,03,970 = 206.75%

New Loss Ratio = Rs 5,21,09,400/ Rs 2,40,03,970 = 217%

Portfolio becomes even more unprofitable.

This indicates that an increase in claim frequency would further worsen portfolio profitability and increase financial risk for the insurer.